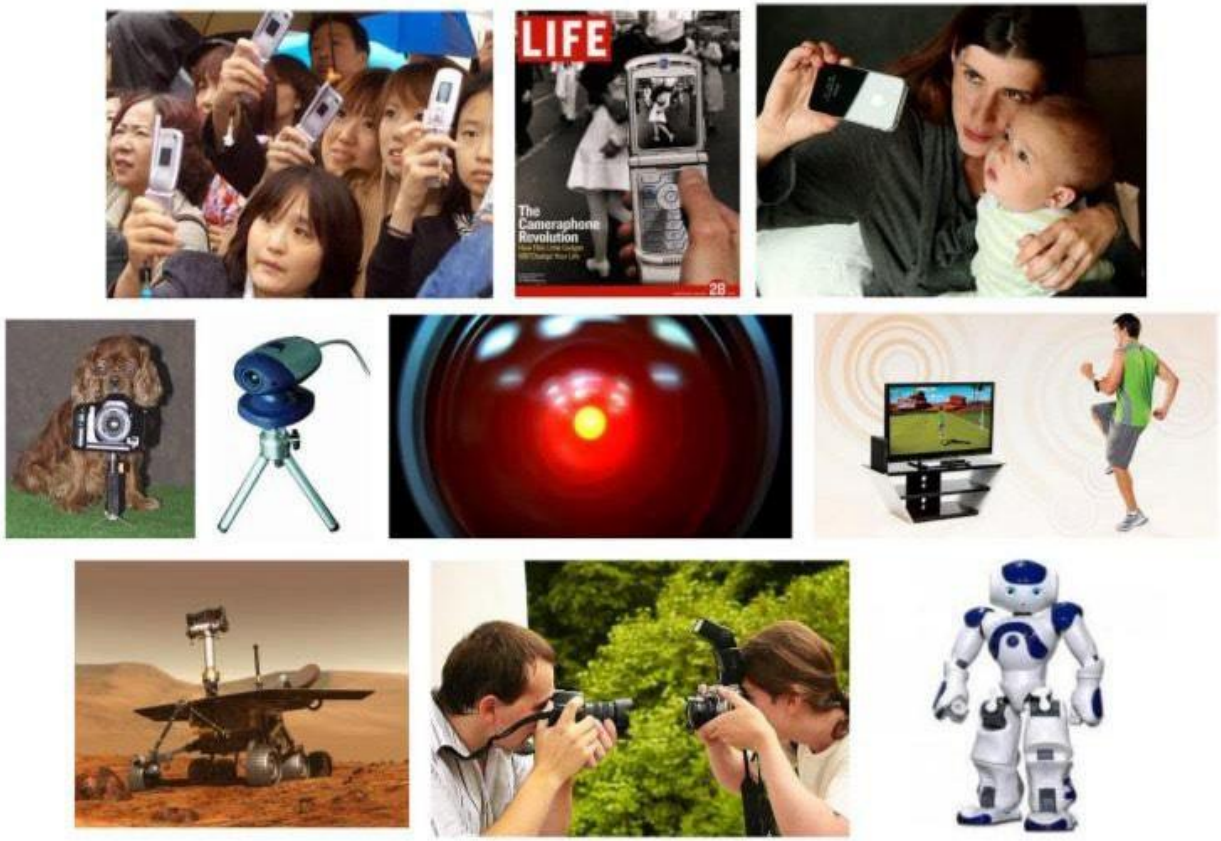
Key Applications of AI: Computer Vision and Natural Language Processing

Dr. Majed Elwardy
25/09/2026

Acknowledgement
These slides are reused **as-is from previous years**, prepared by Dr. Ashir Javeed , DIDA, BTH.

Why study computer vision?

- Computer vision is the science and technology of machines that see and **extract the meaning from the images**.
- It has been used in different **application areas** such as Image Understanding, Security, Machine vision, Robot vision, Image analysis, Video Understanding (and video has sequence of images or lots of pictures) etc.

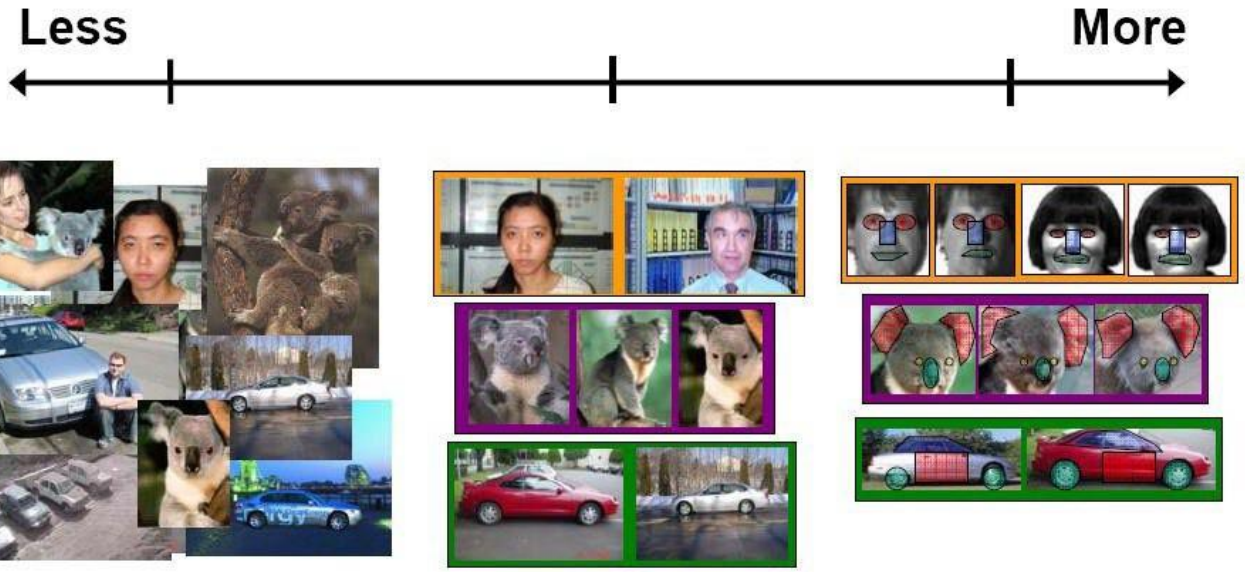


Cont.

- An image is worth 1000 words
- Images and movies are everywhere



Spectrum of supervision



What do you see?



What do you see?



Challenging?

Bad Quality Image

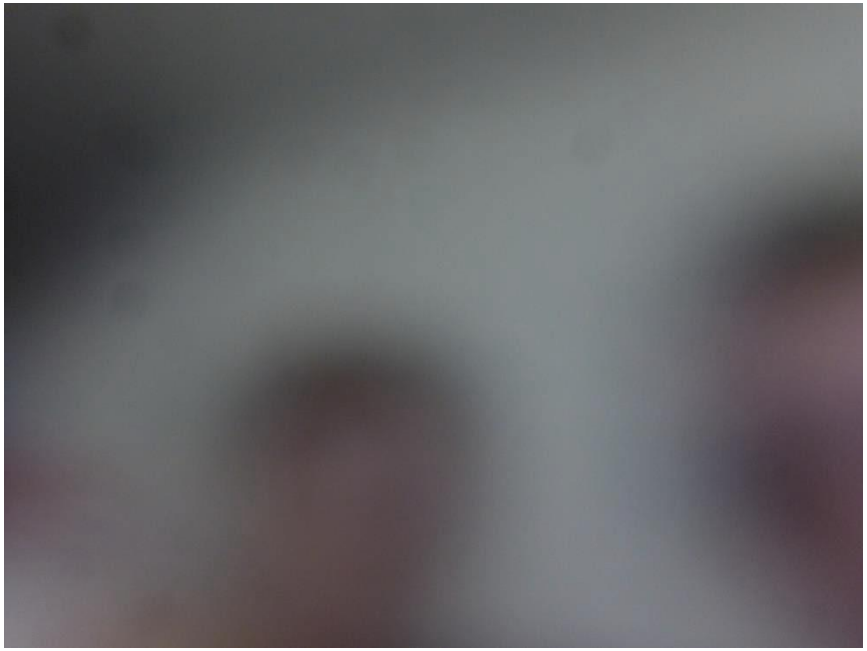
What do you see?



Good Quality Image

The goal of computer vision

To extract "meaning" from pixels



Challenging?

Any suggestion?

What is an image?

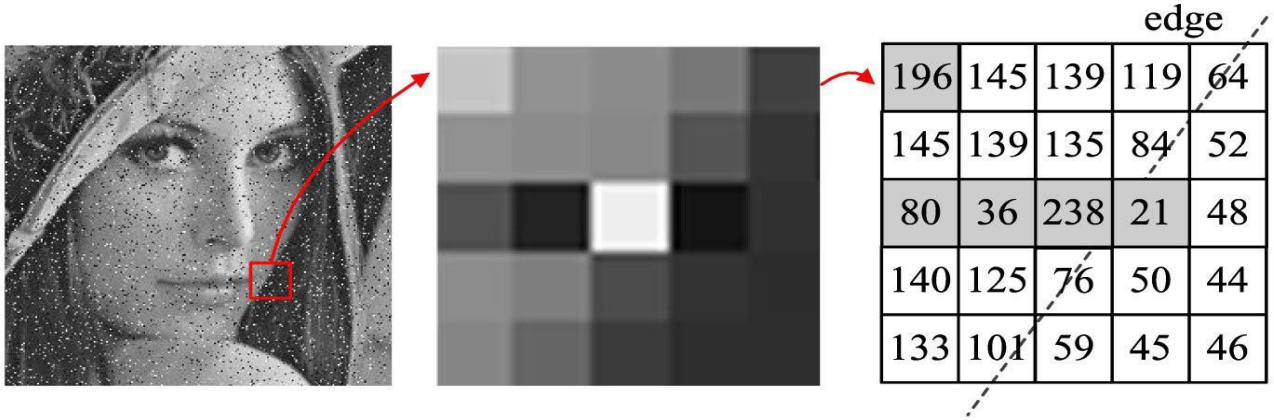
- An image is a 2-dimensional function:

Set of Intensity values

Image coordinates are integers

$$I(x, y) \in \mathbb{P} \quad and \quad x, y \in \mathbb{N}$$

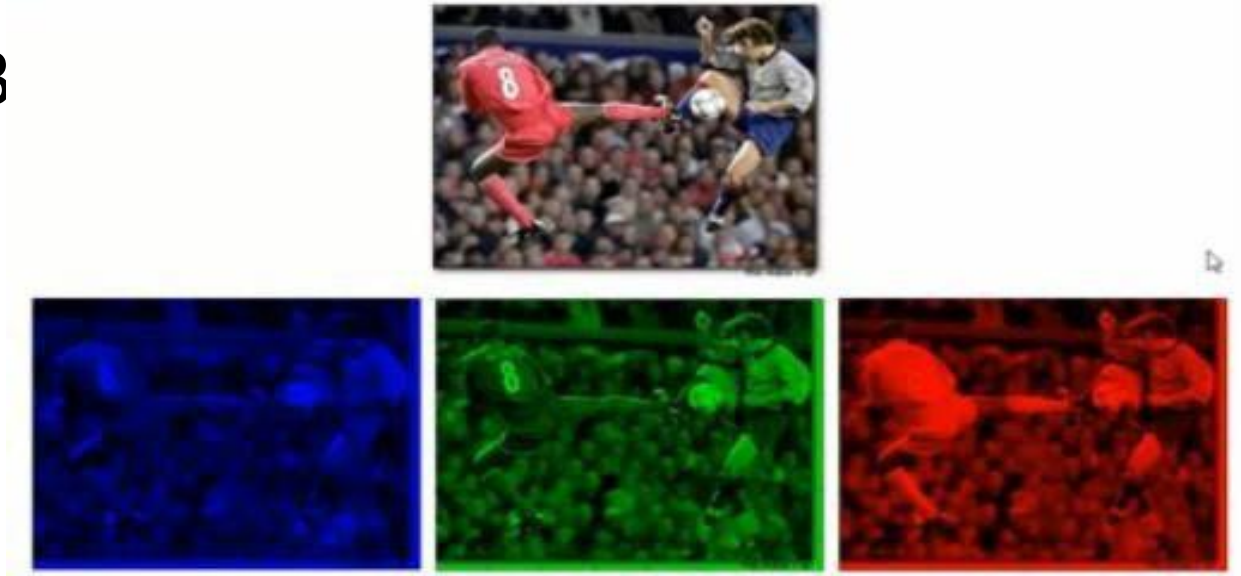
- The amplitude of I is called intensity or gray level at the point (x, y).



(a) original image (b) Region of red rectangle (c) Gray-scale value

Types of image

- RGB





Binary Image

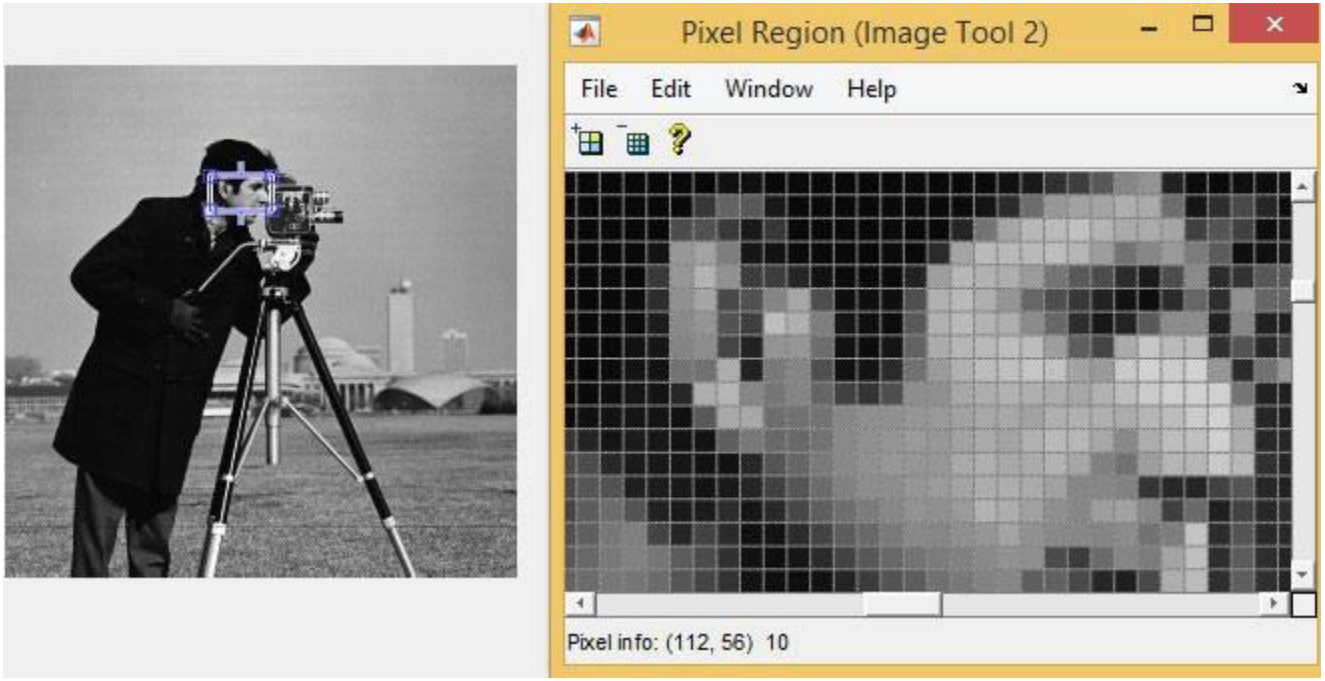


Grayscale



RGB

Example of a Digital Image



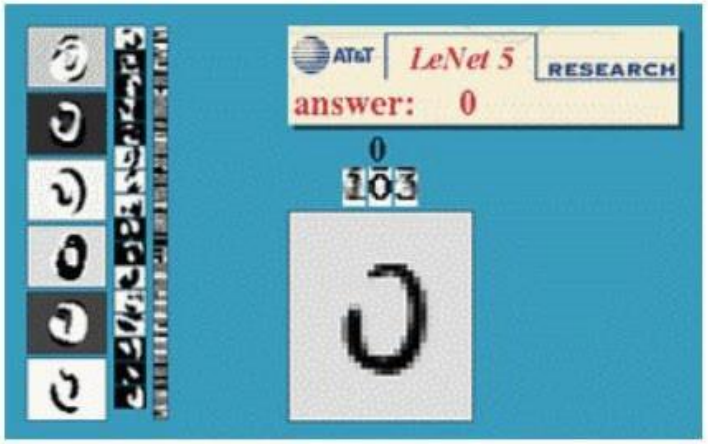
Consider the following image ($255 \times 252\ pixels$) to be 2D function or a matrix with rows and columns

$$I(112,56) = 10$$

x y

Applications

Optical character recognition (OCR)

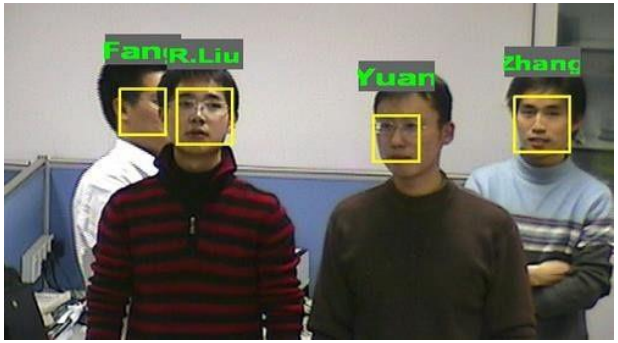


Digit recognition



License plate readers

Face

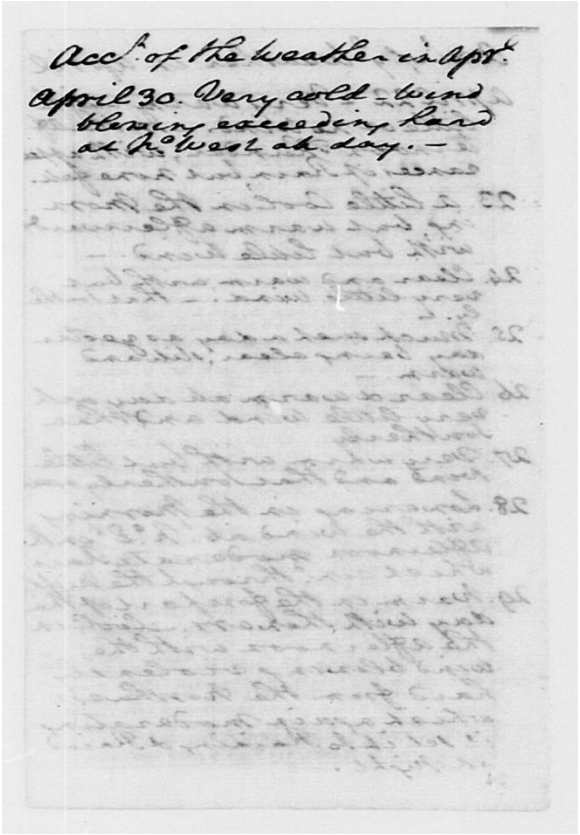




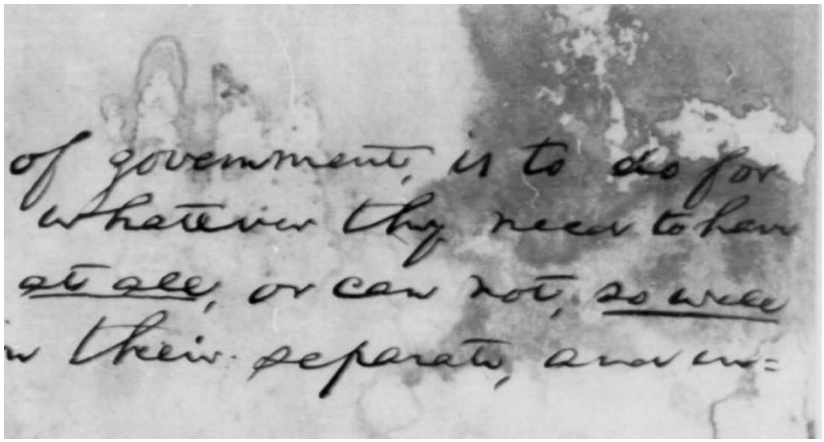
Biometric Recognition

Ear, eyes, Hand, finger etc.

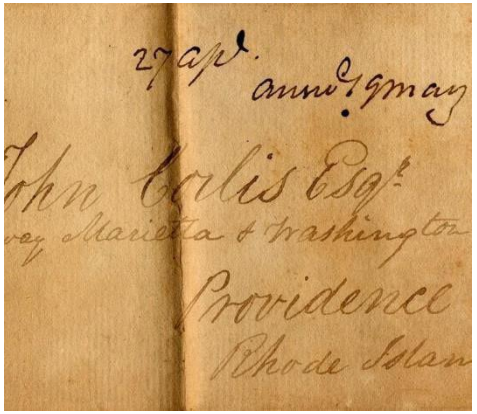
Handwritten Recognition



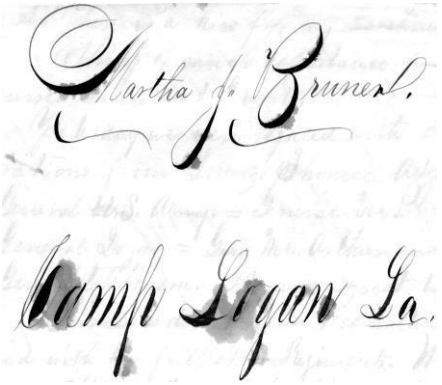
Bleed-through



Bi-model background

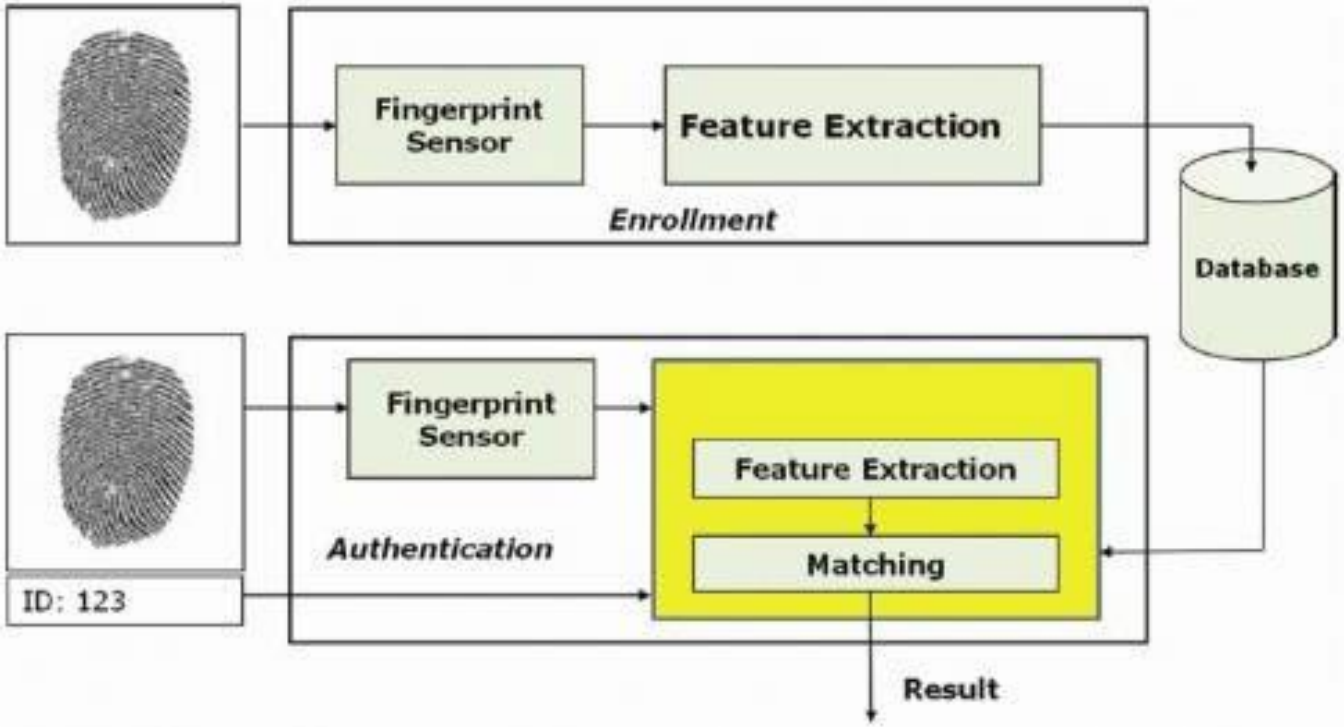


Degradation



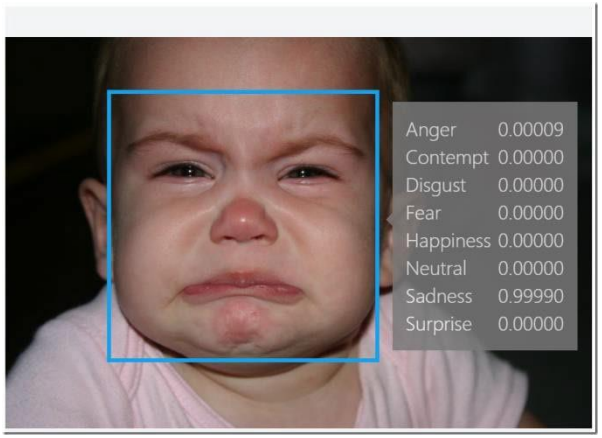
Faint, ink

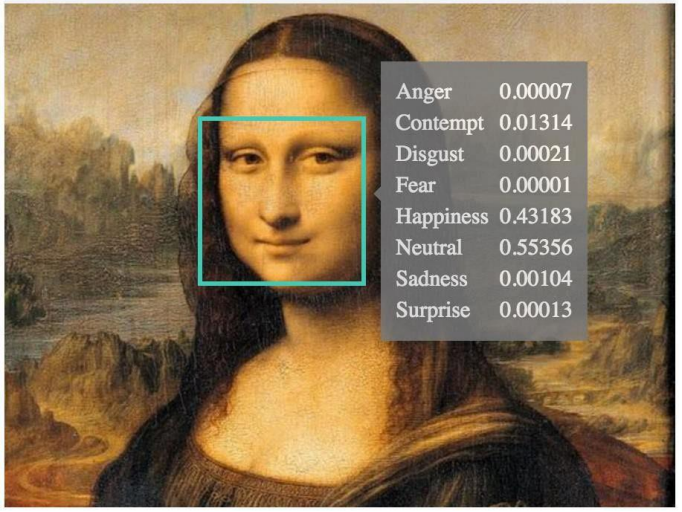
A Block diagram for Fingerprint Recognition



Emotion detection

Sad?

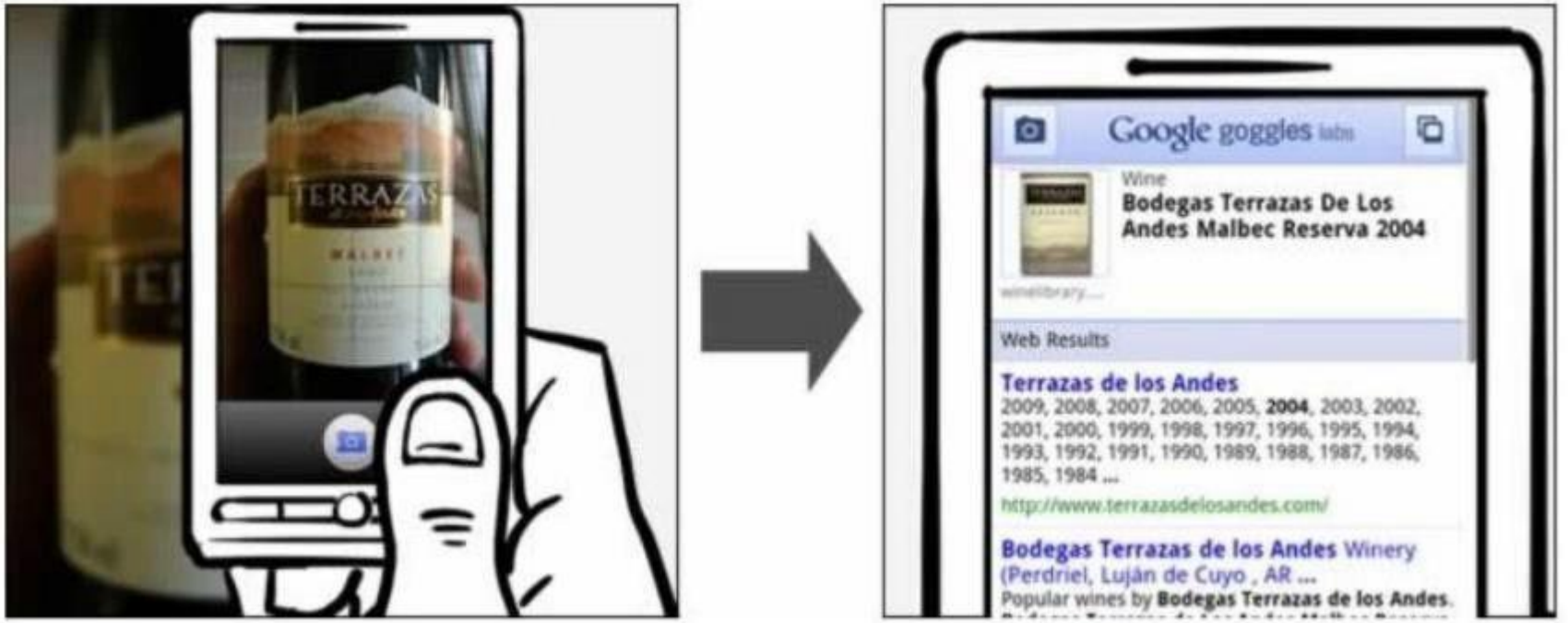




Neutral?



Mobile visual search



Automotive safety



Vision systems in high-end BMW, GM, Volvo models

– Object Detection and Recognition
– Sign Detection and Recognition
– Pedestrian existence warning
– Existence of other cars warning
– Lane distance warning
– Headway monitoring and warning





Self-driving cars



Google car project

Imagine if everyone could get around easily and safely, regardless of their ability to drive.

https://www.google.com/selfdrivingcar/



Tesla
https://www.tesla.com/videos/full-self-driving-hardware-all-tesla-cars?redirect=no

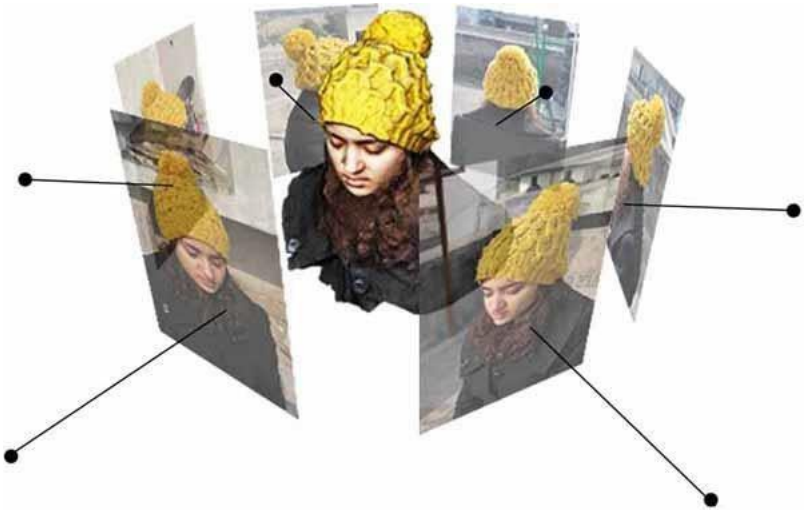
Vision-based interaction: Xbox Kinect

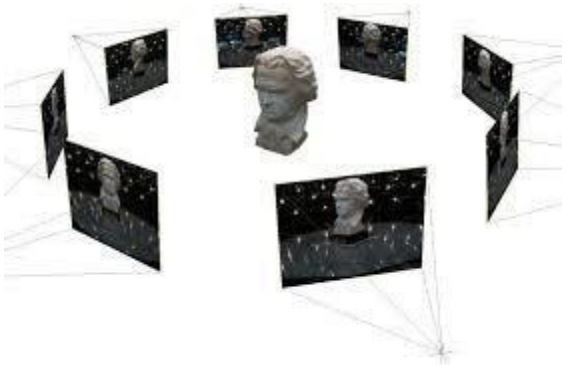






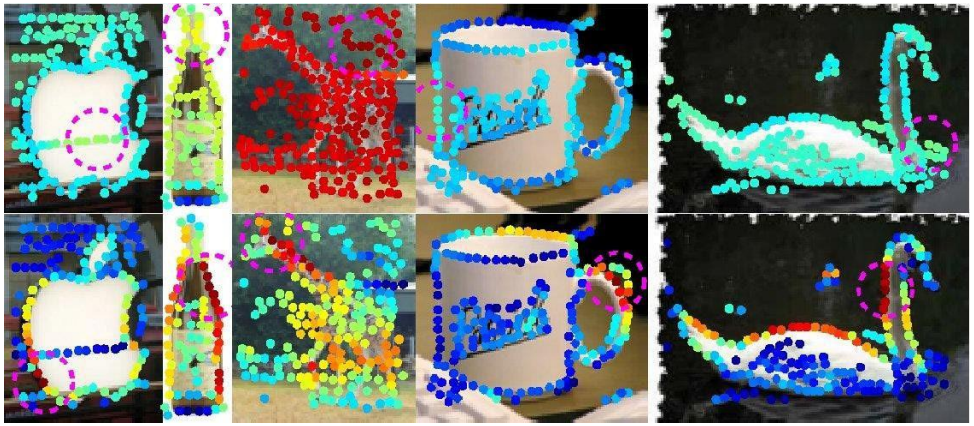
3D Reconstruction







Object recognition

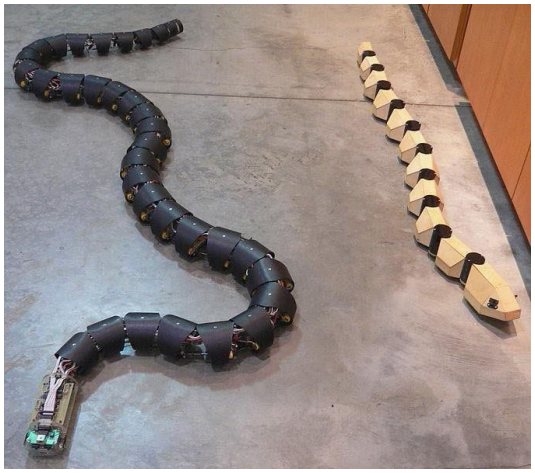








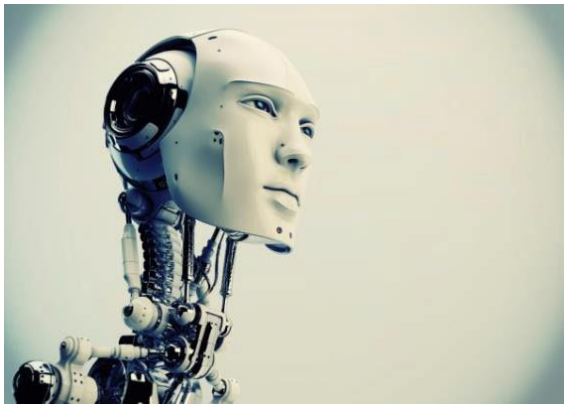
Robotics



Snake



Robocup



Why is computer vision difficult?

- Contrast
- Noise
- Resolution
- Blur
- Motion
- Compression

Contrast



Low Contrast



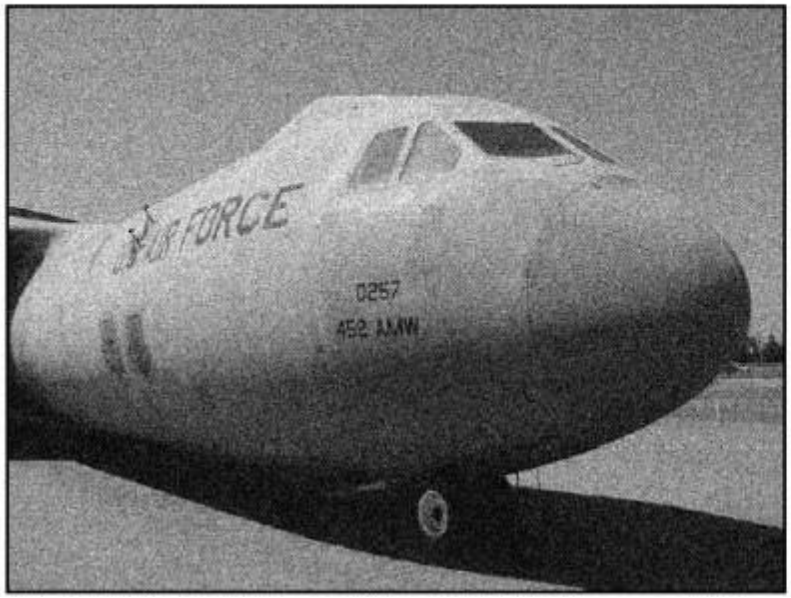
Original Contrast



High Contrast

Noise

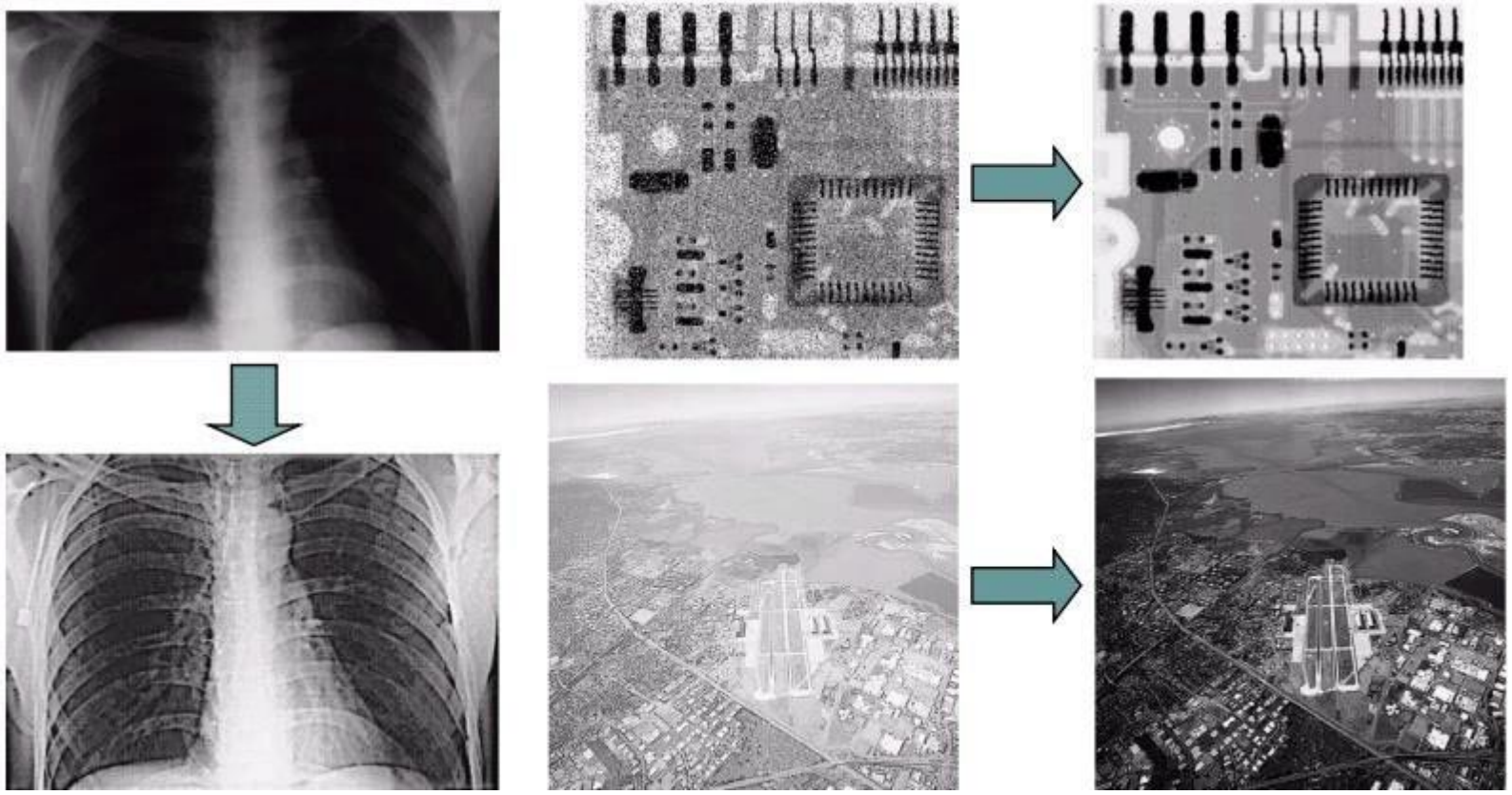
Noisy Image



Denoised Image



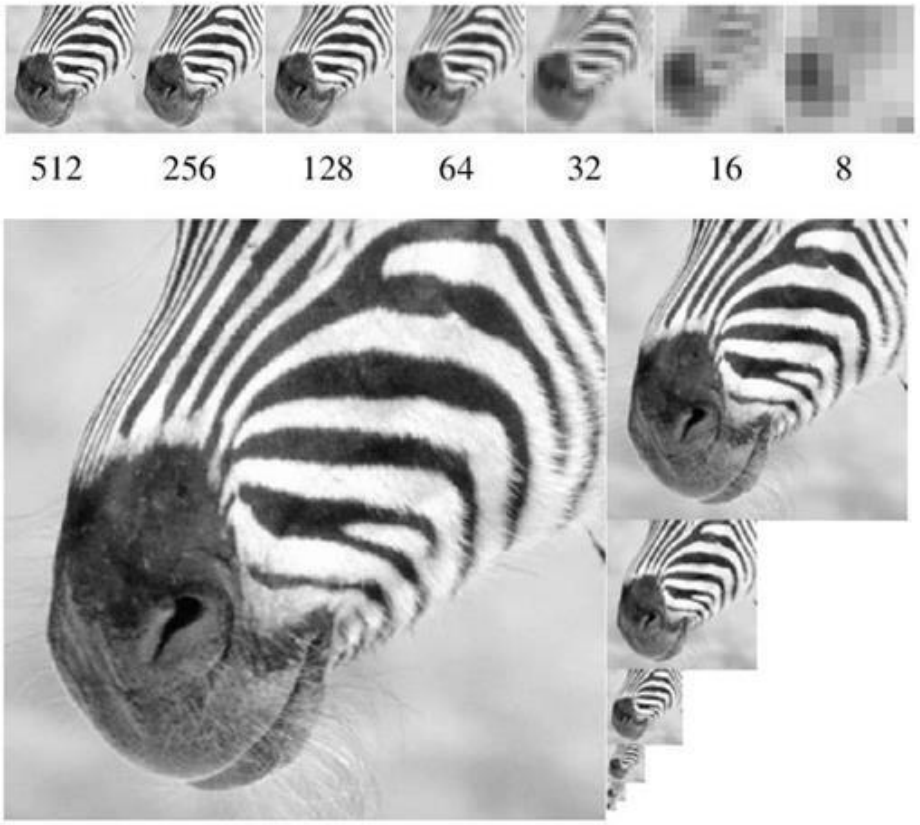
Noise



Blur based on Motion



Scale





The Most Popular Types of Problems in Computer Vision

Image Classification



Is this a dog?

Cont.

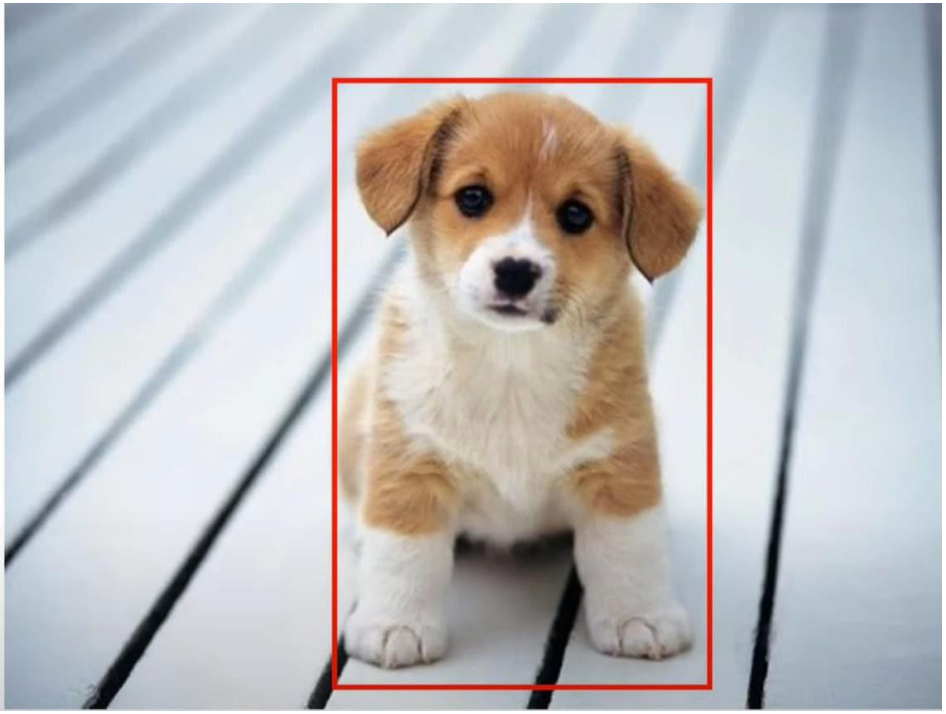
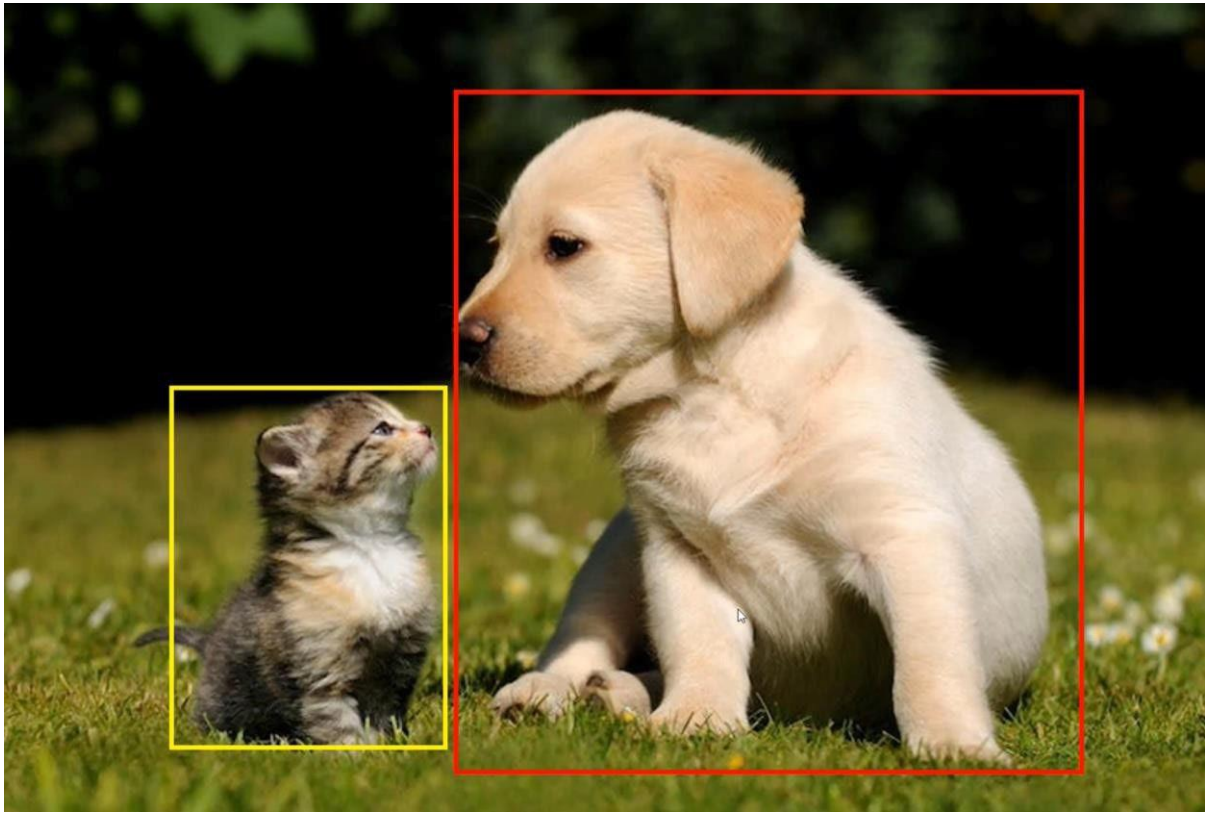


Image Classification with Localization.

Object Detection in images

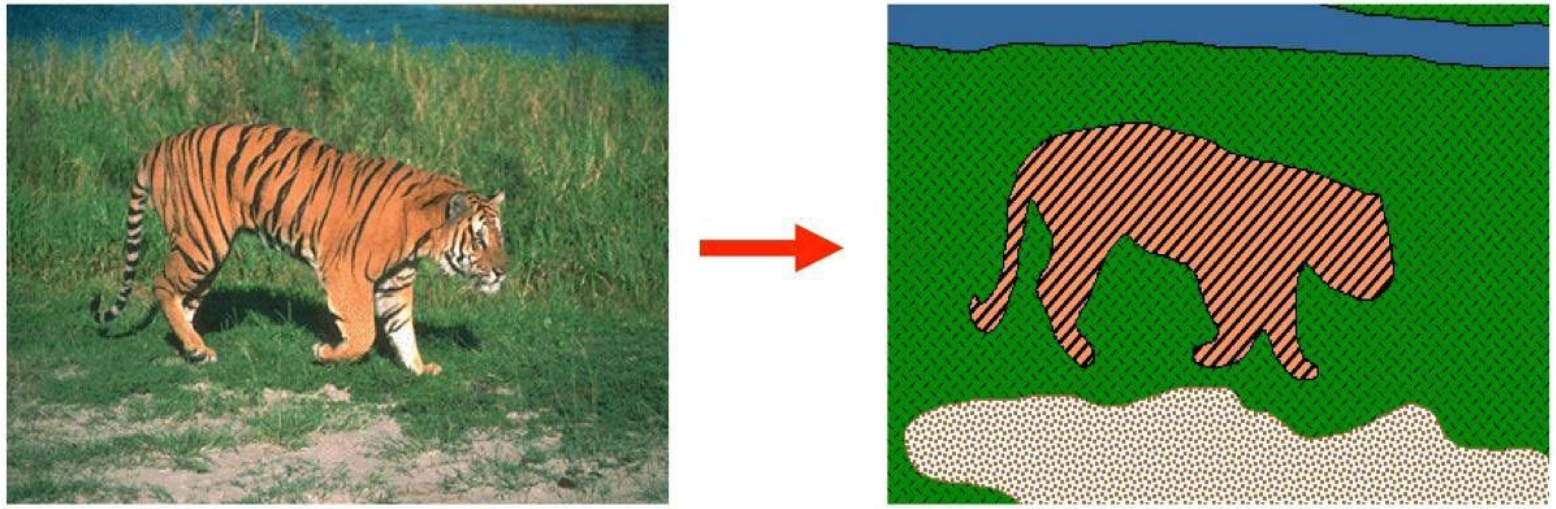


Assume that we have multiple objects in an image. Rectangles are included around each object and localize them. This is called object detection.

Image Segmentation

- Another important subject within computer vision is **image segmentation**.
- It is the process of dividing an image into different regions based on the **characteristics of pixels to identify objects** or boundaries to simplify an image and more efficiently analyze it.
- Segmentation impacts a number of domains, from the film making industry to the field of medicine.
- Image segmentation is also used to track objects in a sequence of images and to classify terrains, like **petroleum reserves**, in **satellite images**.
- Some medical applications of segmentation include the identification of injured muscle, the measurement of bone and tissue, and the detection of suspicious structures to aid radiologists.

Cont.

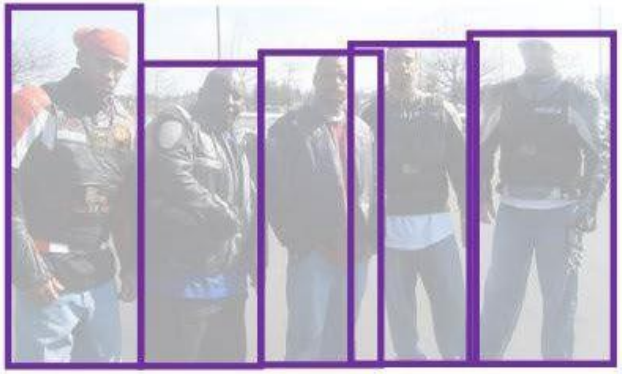


One way to view segmentation is **clustering**, where pixels sharing certain features such as color, intensity, or texture are grouped together and represented as a single entity.

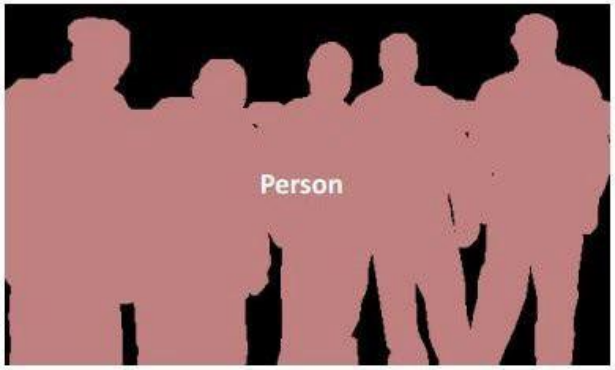
Types of image segmentation

- **Semantic segmentation** is an approach detecting, for every pixel**, belonging class of the object**. For example, when all people in a figure are segmented as one object and background as one object.

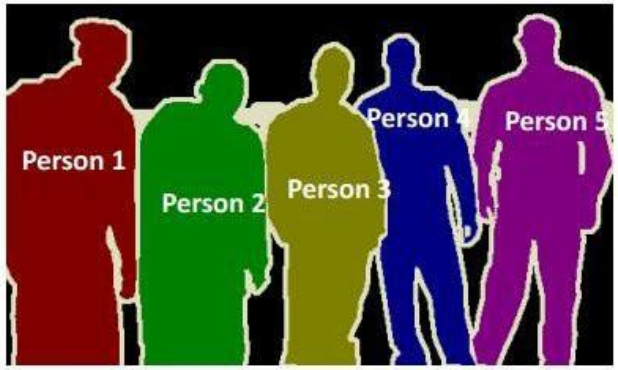
- **Instance segmentation** is an approach that identifies, for every pixel, a **belonging instance of the object**. It detects each distinct object of interest in the image. For example, when each person in a figure is segmented as an individual object.



Object Detection



Semantic Segmentation



Instance Segmentation

Cont.

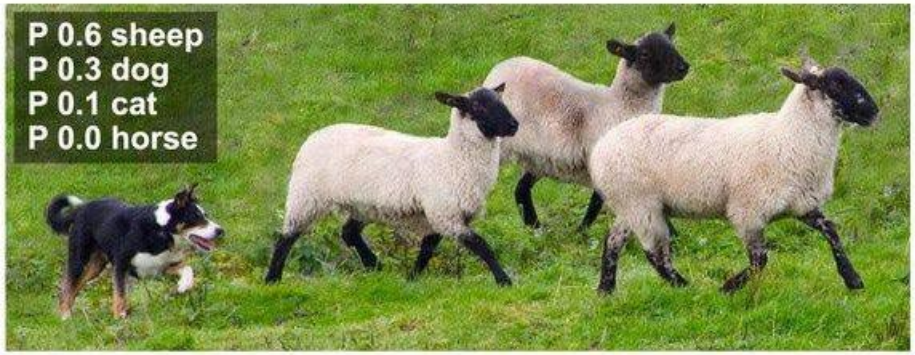
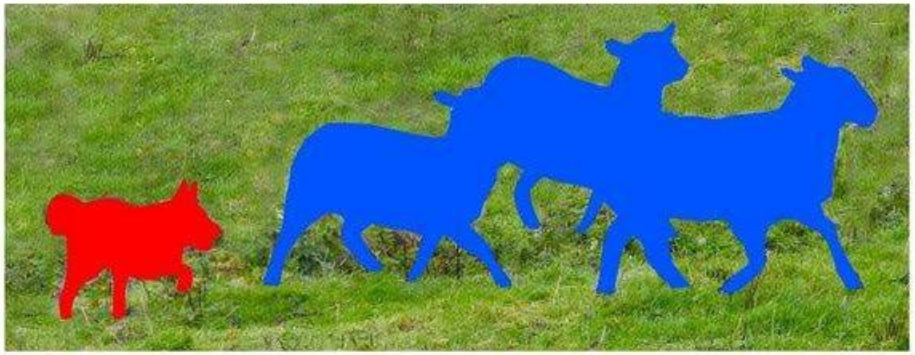
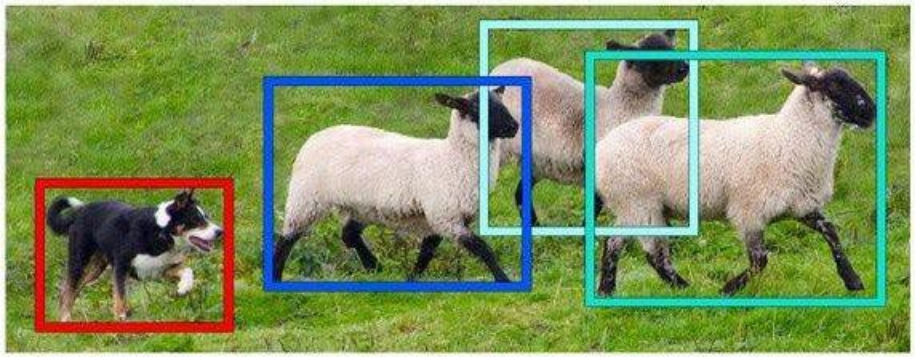


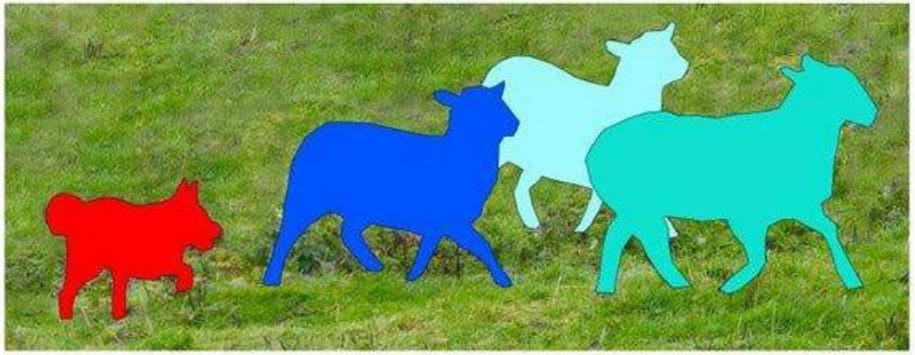
Image Recognition



Semantic Segmentation



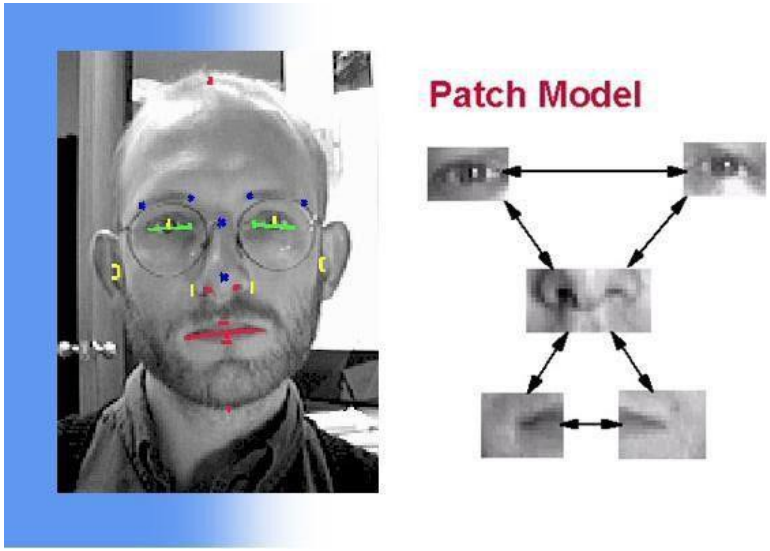
Object Detection



Instance Segmentation

Pattern Recognition

- A pattern is an object, process or event that can be given a name.



Object: Face

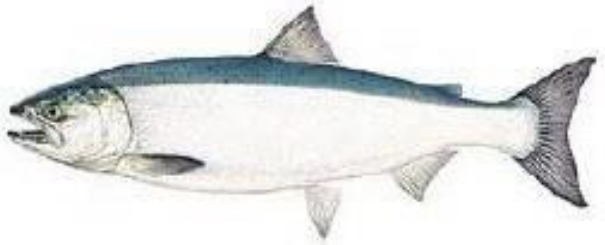


Event: Car accident

Example:



Sea bass



Salmon

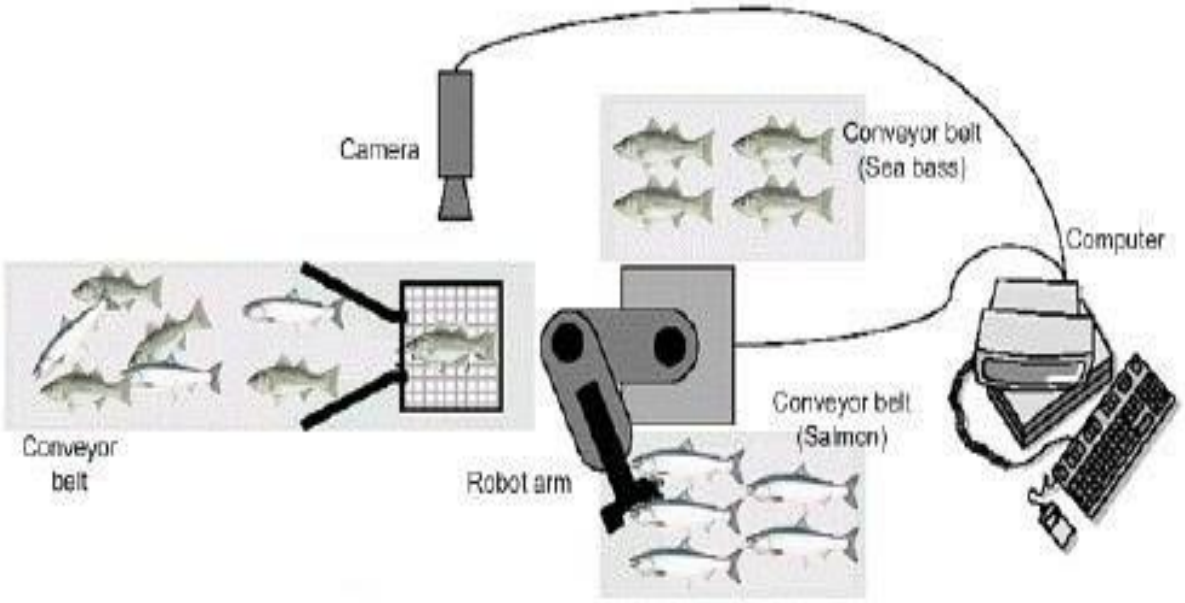


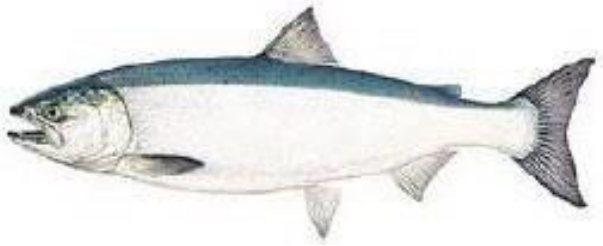
Figure 1.5: The fish-packing system.

- Set up a camera to watch the fish coming through on the conveyor belt.
- Classify each fish as salmon or sea bass.

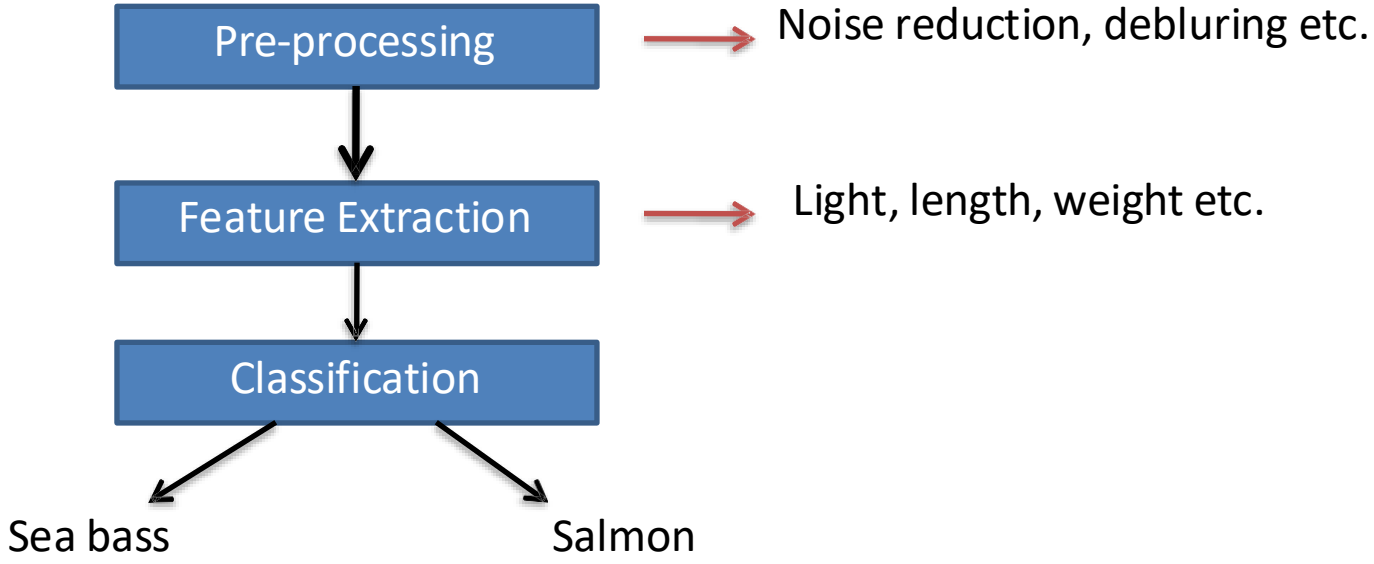
Cont.



Sea bass



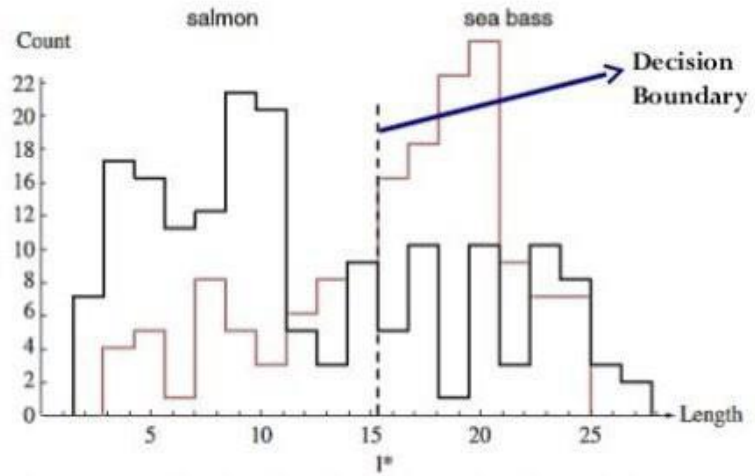
Salmon



Feature: Length

- Suppose an expert at the fish packing plant tells us that length is the best feature.

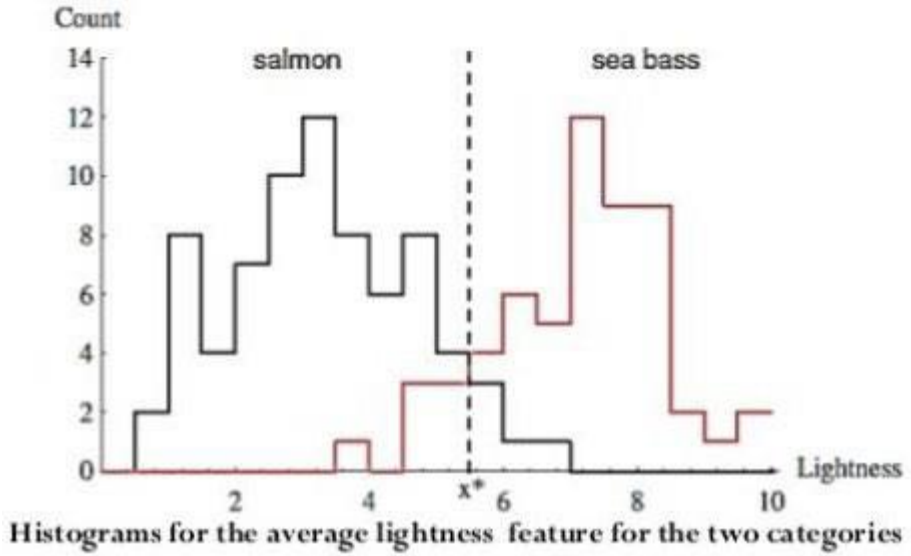
- We trust this expert for the training set.
- Gather a few examples from our installation to analyze the length feature.
- These examples are our training set.
- Want to be sure to gather a representative population of them.
- We analyze the length feature by Building histograms: marginal distributions



Histograms for the length feature for the two categories

Feature: Lightness

- Select the average lightness of the fish as a possible feature for discrimination between two classes.



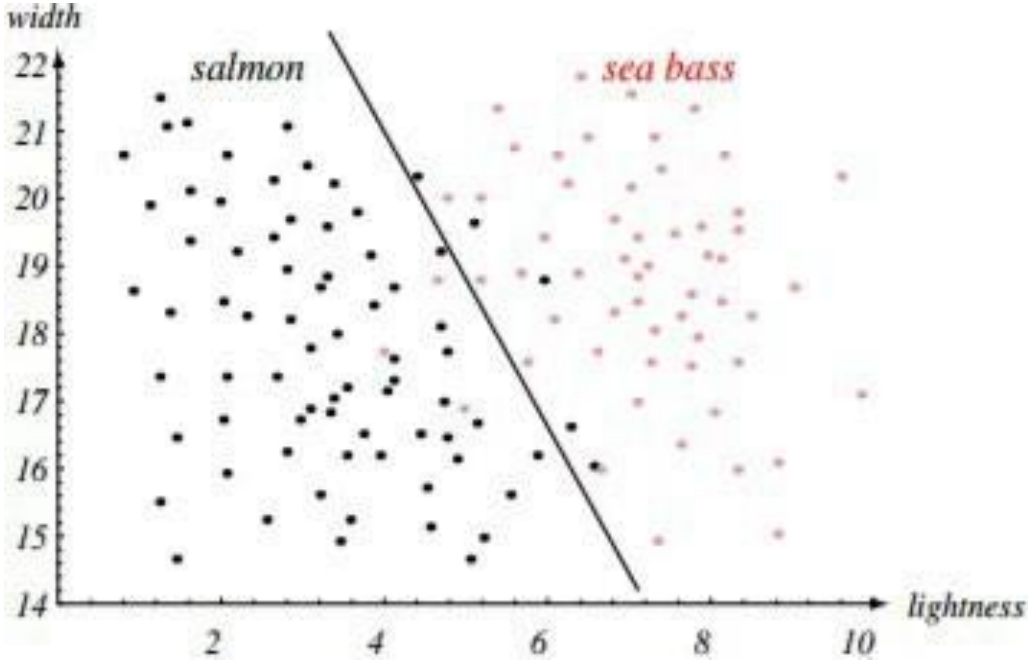
Histograms for the average lightness feature for the two categories

This feature exhibits a much better separation between the two classes.

Features' Combination

- In the fish example, perhaps lightness, x_1, and width, x_2, will jointly do better than any alone. This is an example of a 2D feature space:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

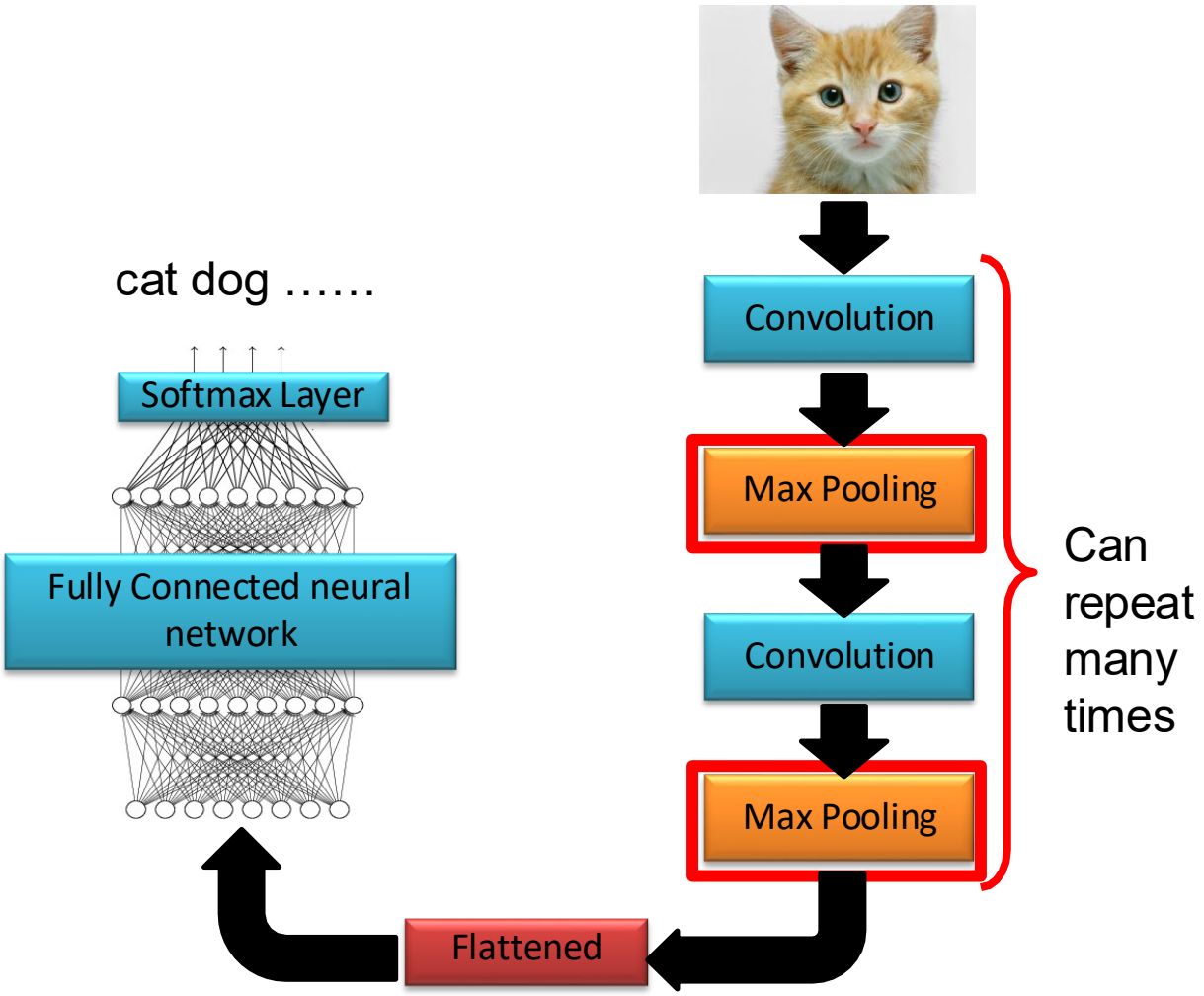


Advantages of Feature' Combination

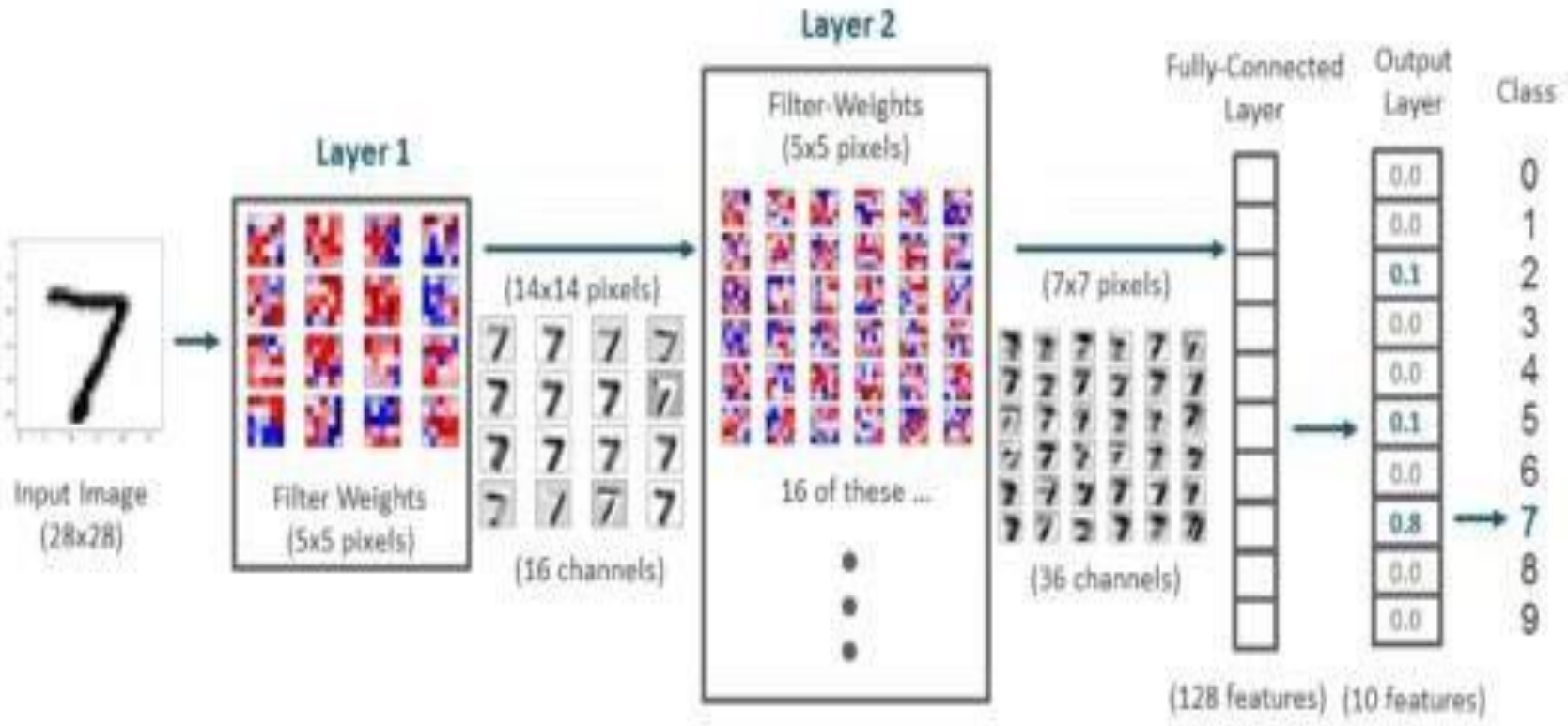
- The two features obviously separate the classes much better than one alone.
- This suggests adding a third feature. And a fourth feature. And so on.

Deep Learning for Computer Vision

The whole CNN



Example


Layer 1
Input Image
(28x28)
Filter Weights
(5x5 pixels)
(14x14 pixels)
(16 channels)
Layer 2
Filter-Weights
(5x5 pixels)
16 of these ...
(7x7 pixels)
(36 channels)
Fully-Connected
Layer
(128 features)
Output
Layer
0.0
0.0
0.1
0.0
0.0
0.1
0.0
0.8
0.0
0.0
(10 features)
Class
0
1
2
3
4
5
6
7
8
9

KEY APPLICATIONS: NATURAL LANGUAGE PROCESSING

Natural Language Processing

Applications

- Machine Translation
- Information Retrieval
- Question Answering
- Dialogue Systems
- Information Extraction
- Summarization
- Sentiment Analysis
- ...

Core Technologies

- Language modeling
- Part-of-speech tagging
- Syntactic parsing
- Named-entity recognition
- Word sense disambiguation
- Semantic role labeling
- ...

NLP lies at the intersection of computational linguistics and machine learning.

Words

- Language Modeling
  - Statistical Language Modeling
  - Neural Language Modeling
- Spelling correction

Dictionary-based

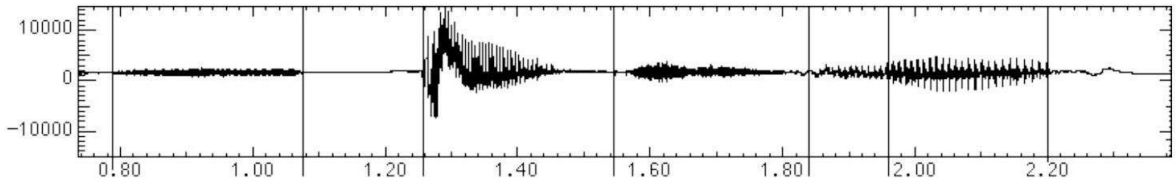
- Input: `speling`
- Closest dictionary match using edit distance: `spelling`.

Context-aware

- Input: `I went to the store to by milk`
- Model predicts: `buy` instead of `by`.

Speech Systems

- Automatic Speech Recognition (ASR)
  - Audio in, text out
    - Example: Virtual assistants like Siri, Alexa, or Google Assistant





"Speech Lab"

- Text to Speech (TTS)
  - Text in, audio out
  - Example: Screen readers, GPS navigation systems.



Example: Siri

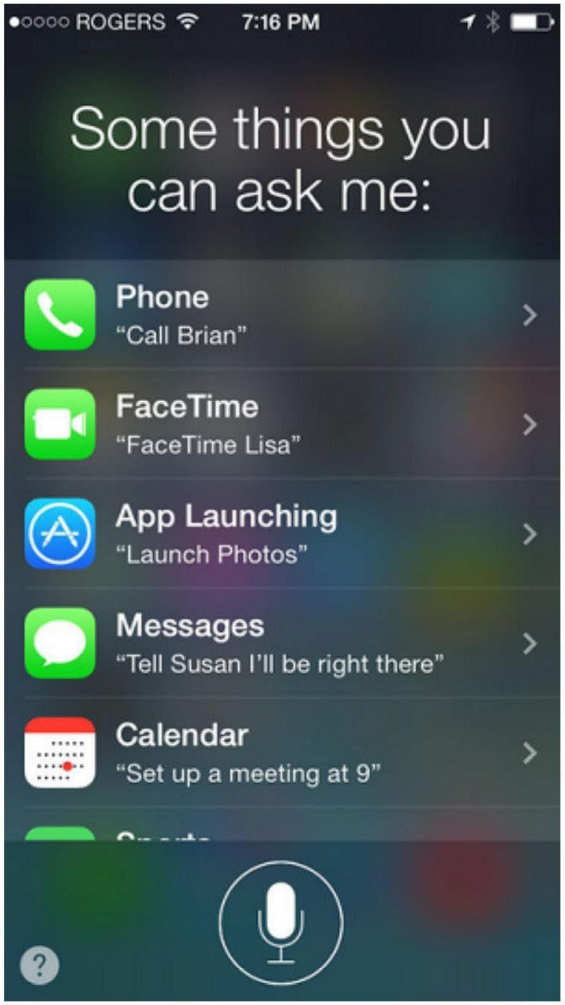
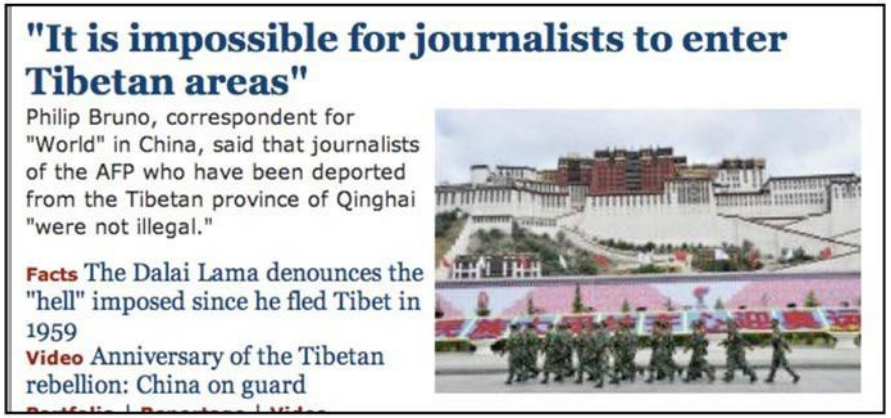


Image: Wikipedia

- Siri contains
  - Speech recognition
  - Language analysis
  - Dialog processing
  - Text to speech

Machine Translation





a computer system converts text or speech in a source language (e.g., English) into a target language (e.g., French) while preserving meaning, grammar, and style.

Goal: Achieve accurate, fluent, and contextually appropriate translations without human intervention

Challenges in Machine Translation

- Ambiguity

  Example: "bank" → financial institution or river bank

- Idiomatic Expressions

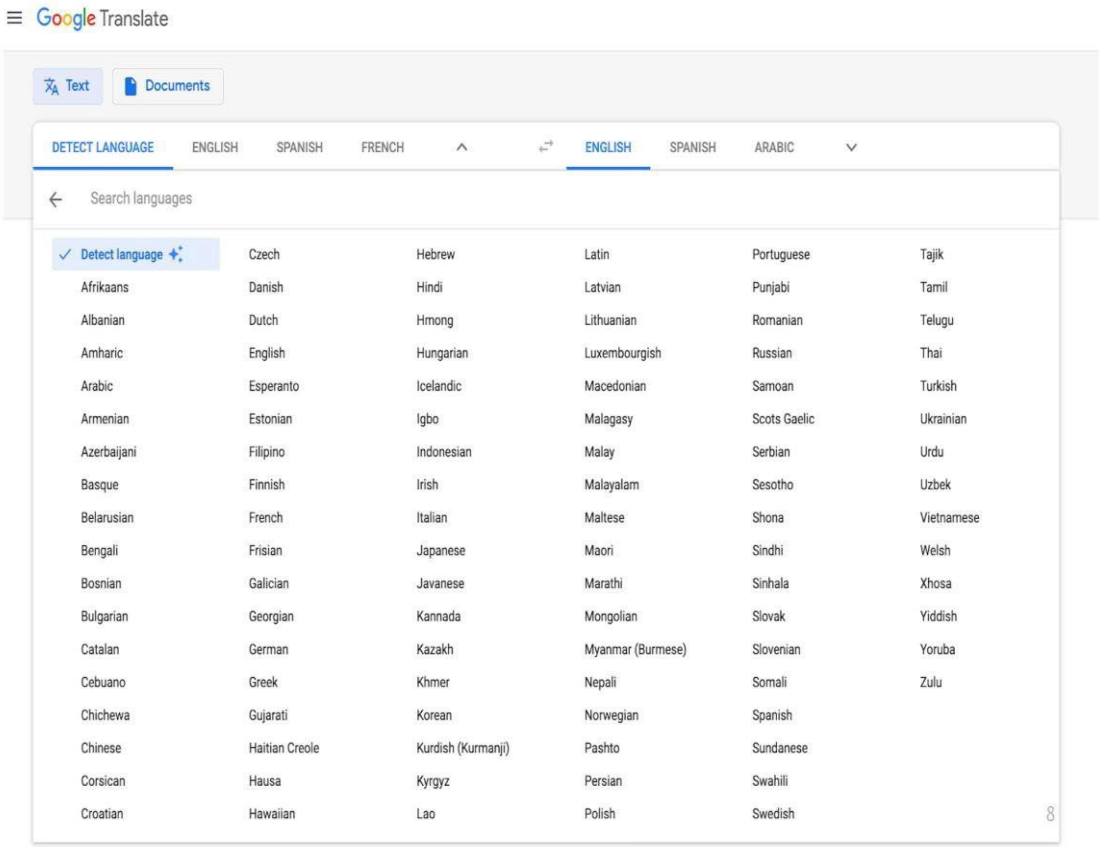
- Context Understanding

  Example: "He saw her duck" – is it a bird or the action?

- Morphological Complexity

Cont.


Google Translate
CHINESE - DETECTED
ENGLISH
我学习深度学习和机器学习
Wǒ xuéxí shēndù xuéxí hé jīqì xuéxí
I study deep learning and machine learning.
Send feedback


Google Translate
Text
Documents
DETECT LANGUAGE
ENGLISH
SPANISH
FRENCH
ENGLISH
SPANISH
ARABIC
Search languages
Detect language
Afrikaans
Albanian
Amharic
Arabic
Armenian
Azerbaijani
Basque
Belarusian
Bengali
Bosnian
Bulgarian
Catalan
Cebuano
Chichewa
Chinese
Corsican
Croatian
Czech
Danish
Dutch
English
Esperanto
Estonian
Filipino
Finnish
French
Frisian
Galician
Georgian
German
Greek
Gujarati
Haitian Creole
Hausa
Hawaiian
Hebrew
Hindi
Hmong
Hungarian
Icelandic
Igbo
Indonesian
Irish
Italian
Japanese
Javanese
Kannada
Kazakh
Khmer
Korean
Kurdish (Kurmanji)
Kyrgyz
Lao
Latin
Latvian
Lithuanian
Luxembourgish
Macedonian
Malagasy
Malay
Malayalam
Maltese
Maori
Marathi
Mongolian
Myanmar (Burmese)
Nepali
Norwegian
Pashto
Persian
Polish
Portuguese
Punjabi
Romanian
Russian
Samoan
Scots Gaelic
Serbian
Sesotho
Shona
Sindhi
Sinhala
Slovak
Slovenian
Somali
Spanish
Sundanese
Swahili
Swedish
Tajik
Tamil
Telugu
Thai
Turkish
Ukrainian
Urdu
Uzbek
Vietnamese
Welsh
Xhosa
Yiddish
Yoruba
Zulu

Deeper Understanding: Reference

Q: Who signed the Serve America Act?

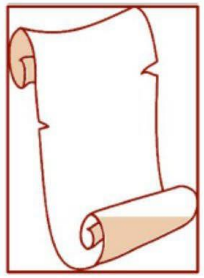
A: Barack Obama



Los Angeles Times

President Barack Obama received the Serve America Act after congress' vote. He signed the bill last Thursday. The president said it would greatly increase service opportunities for the American people.

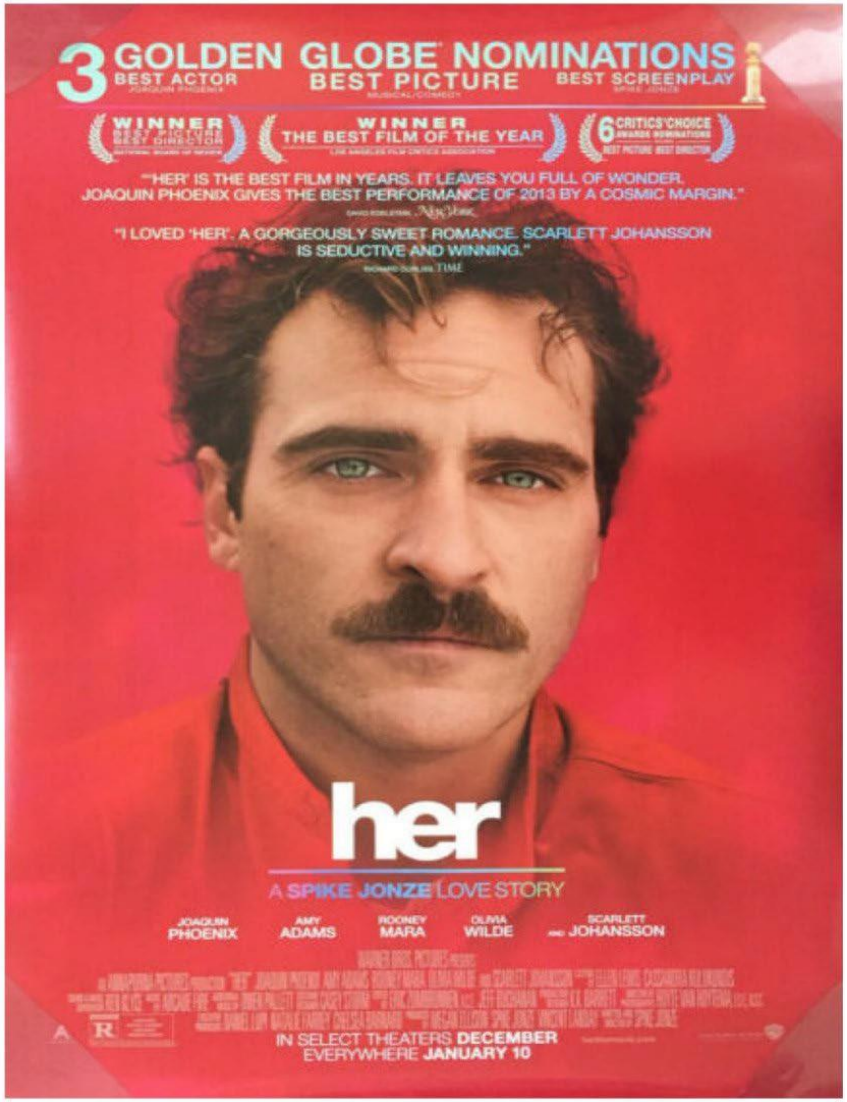
Cont.



President Barack Obama received the Serve America Act after congress' vote. He signed the bill last Thursday. The president said it would greatly increase service opportunities for the American people.

Movies about NLP



Why NLP is Hard?

1. Ambiguity
2. Scale
3. Sparsity
4. Variation
5. Expressivity
6. Unmodeled Variables
7. Unknown representations

Ambiguity

- Ambiguity at multiple levels
  - Word senses: **bank** (finance or river ?)
  - Part of speech: **chair** (noun or verb ?)

Sparsity

- Sparse data due to Zipf's Law
- Example: the frequency of different words in a large text corpus

any word	
Frequency	Token
1,698,599	the
849,256	of
793,731	to
640,257	and
508,560	in
407,638	that
400,467	is
394,778	a
263,040	I

nouns	
Frequency	Token
124,598	European
104,325	Mr
92,195	Commission
66,781	President
62,867	Parliament
57,804	Union
53,683	report
53,547	Council
45,842	States

Expressivity

- Not only can one form have different meanings (ambiguity) but the same meaning can be expressed with different forms:
  - *She gave the book to Tom* vs. *She gave Tom the book*
  - *Some kids popped by* vs. *A few children visited*
  - *Is that window still open?* vs. *Please close the window*

Fields with Connections to NLP

- Machine learning
- Linguistics (including psycho-, socio-, descriptive, and theoretical)
- Cognitive science
- Information theory
- Logic
- Data science
- Political science
- Psychology
- Economics
- Education

Today's Applications

- Conversational agents
- Information extraction and question answering
- Machine translation
- Opinion and sentiment analysis
- Social media analysis
- Visual understanding
- Essay evaluation
- Mining legal, medical, or scholarly literature